

LAB DAY 4

NAME: AGILAN VIYASH S

REG. NO.: 192511346

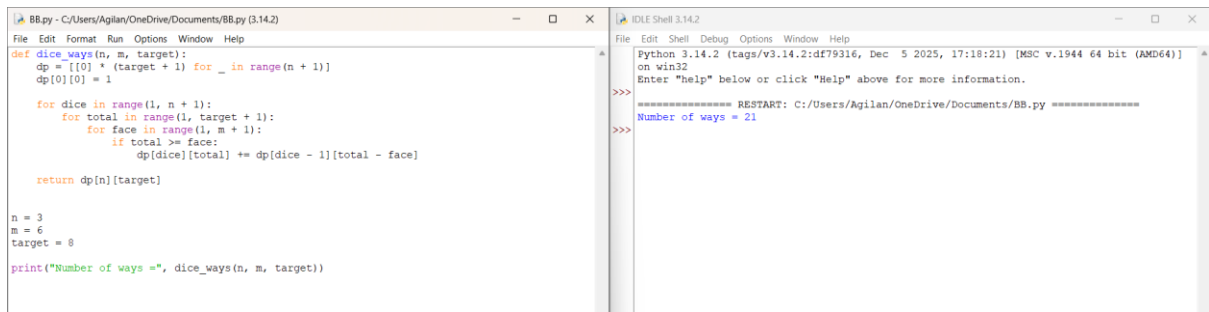
COURSE CODE AND NAME: CSA0617 DESIGN AND ANALYSIS OF ALGORITHM

DICE THROW PROBLEM

Q1. 3 DICE, 6 FACES, TARGET=8:

```
def dice_ways(n, m, target):  
    dp = [[0] * (target + 1) for _ in range(n + 1)]  
    dp[0][0] = 1  
  
    for dice in range(1, n + 1):  
        for total in range(1, target + 1):  
            for face in range(1, m + 1):  
                if total >= face:  
                    dp[dice][total] += dp[dice - 1][total - face]  
  
    return dp[n][target]  
  
n = 3  
m = 6  
target = 8  
  
print("Number of ways =", dice_ways(n, m, target))
```

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following code:

```
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for dice in range(1, n + 1):
        for total in range(1, target + 1):
            for face in range(1, m + 1):
                if total >= face:
                    dp[dice][total] += dp[dice - 1][total - face]

    return dp[n][target]

n = 3
m = 6
target = 8

print("Number of ways =", dice_ways(n, m, target))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 21
>>>
```

Q2. 2 DICE, 6 FACES, TARGET=7 :

def dice_ways(n, m, target):

dp = [[0] * (target + 1) for _ in range(n + 1)]

dp[0][0] = 1

for i in range(1, n + 1):

for j in range(1, target + 1):

for face in range(1, m + 1):

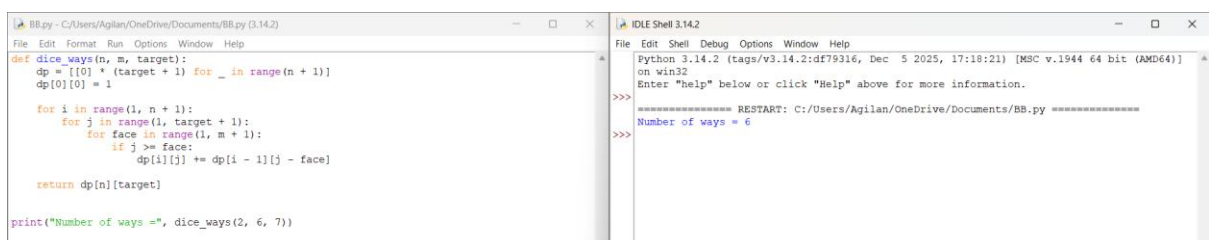
if j >= face:

dp[i][j] += dp[i - 1][j - face]

return dp[n][target]

print("Number of ways =", dice_ways(2, 6, 7))

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following code:

```
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for j in range(1, target + 1):
            for face in range(1, m + 1):
                if j >= face:
                    dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(2, 6, 7))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 6
>>>
```

Q3. 4 DICE, 4 FACES, TAGRET=10:

```

def dice_ways(n, m, target):

    dp = [[0] * (target + 1) for _ in range(n + 1)]

    dp[0][0] = 1

    for i in range(1, n + 1):

        for j in range(1, target + 1):

            for face in range(1, m + 1):

                if j >= face:

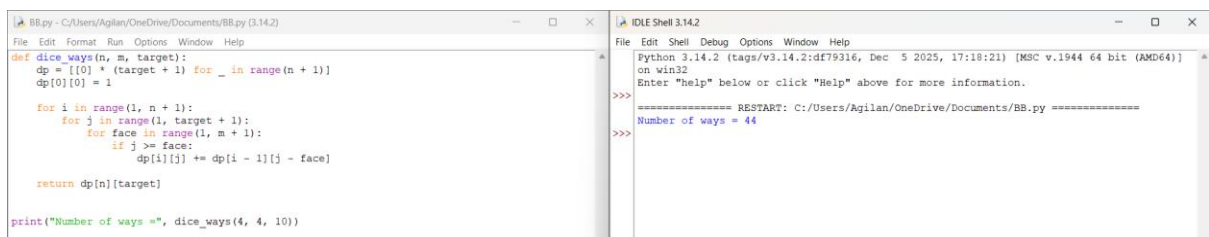
                    dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(4, 4, 10))

```

EXECUTION & OUTPUT:



```

def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for j in range(1, target + 1):
            for face in range(1, m + 1):
                if j >= face:
                    dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(4, 4, 10))

```

```

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 44
>>>

```

Q4. 3 DICE, 6 FACES, TARGET=12:

```

def dice_ways(n, m, target):

    dp = [[0] * (target + 1) for _ in range(n + 1)]

    dp[0][0] = 1

    for i in range(1, n + 1):

        for j in range(1, target + 1):

            for face in range(1, m + 1):

```

```

if j >= face:
    dp[i][j] += dp[i - 1][j - face]

```

```

return dp[n][target]

```

```

print("Number of ways =", dice_ways(3, 6, 12))

```

EXECUTION & OUTPUT:

```

def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for j in range(1, target + 1):
            for face in range(1, m + 1):
                if j >= face:
                    dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(3, 6, 12))

```

```

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 25
>>>

```

Q5. 5 DICE, 6 FACES, TARGET=15:

```

def dice_ways(n, m, target):

```

```

    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

```

```

    for i in range(1, n + 1):

```

```

        for j in range(1, target + 1):

```

```

            for face in range(1, m + 1):

```

```

                if j >= face:

```

```

                    dp[i][j] += dp[i - 1][j - face]

```

```

    return dp[n][target]

```

```

print("Number of ways =", dice_ways(5, 6, 15))

```

EXECUTION & OUTPUT:

The screenshot shows a Python IDE with two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following code:

```
def dice_ways(n, m, target):  
    dp = [[0] * (target + 1) for _ in range(n + 1)]  
    dp[0][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, target + 1):  
            for face in range(1, m + 1):  
                if j >= face:  
                    dp[i][j] += dp[i - 1][j - face]  
  
    return dp[n][target]  
  
print("Number of ways =", dice_ways(6, 15))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
>>>  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====  
Number of ways = 651  
>>>
```

Q6. 3 DICE, 5 FACES, TARGET=9:

def dice_ways(n, m, target):

dp = [[0] * (target + 1) for _ in range(n + 1)]

dp[0][0] = 1

for i in range(1, n + 1):

for j in range(1, target + 1):

for face in range(1, m + 1):

if j >= face:

dp[i][j] += dp[i - 1][j - face]

return dp[n][target]

print("Number of ways =", dice_ways(3, 5, 9))

EXECUTION & OUTPUT:

The screenshot shows a Python IDE with two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following code:

```
def dice_ways(n, m, target):  
    dp = [[0] * (target + 1) for _ in range(n + 1)]  
    dp[0][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, target + 1):  
            for face in range(1, m + 1):  
                if j >= face:  
                    dp[i][j] += dp[i - 1][j - face]  
  
    return dp[n][target]  
  
print("Number of ways =", dice_ways(3, 5, 9))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

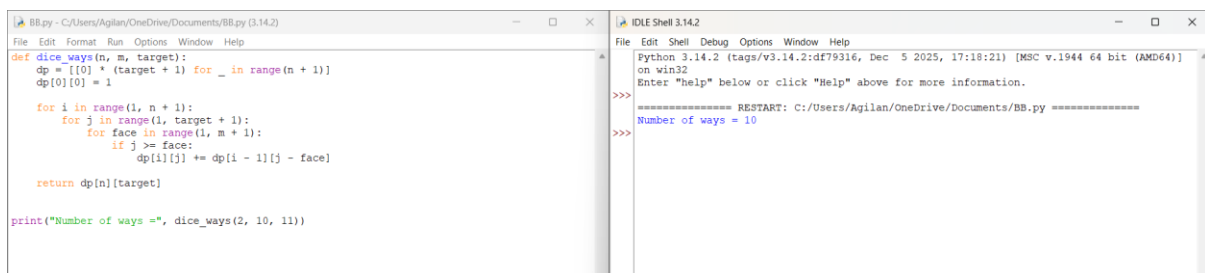
```
>>>  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====  
Number of ways = 19  
>>>
```

Q7. 2 DICE, 10 FACES, TARGET=11:

```
def dice_ways(n, m, target):  
    dp = [[0] * (target + 1) for _ in range(n + 1)]  
    dp[0][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, target + 1):  
            for face in range(1, m + 1):  
                if j >= face:  
                    dp[i][j] += dp[i - 1][j - face]  
  
    return dp[n][target]
```

```
print("Number of ways =", dice_ways(2, 10, 11))
```

EXECUTION & OUTPUT:



```
BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)  
File Edit Format Run Options Window Help  
def dice_ways(n, m, target):  
    dp = [[0] * (target + 1) for _ in range(n + 1)]  
    dp[0][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, target + 1):  
            for face in range(1, m + 1):  
                if j >= face:  
                    dp[i][j] += dp[i - 1][j - face]  
  
    return dp[n][target]  
  
print("Number of ways =", dice_ways(2, 10, 11))  
  
IDLE Shell 3.14.2  
File Edit Shell Debug Options Window Help  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====  
Number of ways = 10  
>>>
```

Q8. GNEREAL N DICE, M FACES:

```
def dice_ways(n, m, target):  
    dp = [[0] * (target + 1) for _ in range(n + 1)]  
    dp[0][0] = 1
```

```

for i in range(1, n + 1):
    for total in range(1, target + 1):
        for face in range(1, m + 1):
            if total >= face:
                dp[i][total] += dp[i - 1][total - face]

return dp[n][target]

```

n = 2

m = 6

target = 7

```
print("Number of ways =", dice_ways(n, m, target))
```

EXECUTION & OUTPUT:

```

BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)
File Edit Format Run Options Window Help
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1
    for i in range(1, n + 1):
        for total in range(1, target + 1):
            for face in range(1, m + 1):
                if total >= face:
                    dp[i][total] += dp[i - 1][total - face]
    return dp[n][target]

n = 2
m = 6
target = 7
print("Number of ways =", dice_ways(n, m, target))

IDLE Shell 3.14.2
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
>>>
Number of ways = 6
>>>

```

Q9. 3 DICE, 6 FACES, TARGET=8:

```
def dice_ways(n, m, target):
```

```
    dp = [[0] * (target + 1) for _ in range(n + 1)]
```

```
    dp[0][0] = 1
```

```
    for i in range(1, n + 1):
```

```
        for total in range(1, target + 1):
```

```

for face in range(1, m + 1):

    if total >= face:

        dp[i][total] += dp[i - 1][total - face]

```

```

return dp[n][target]

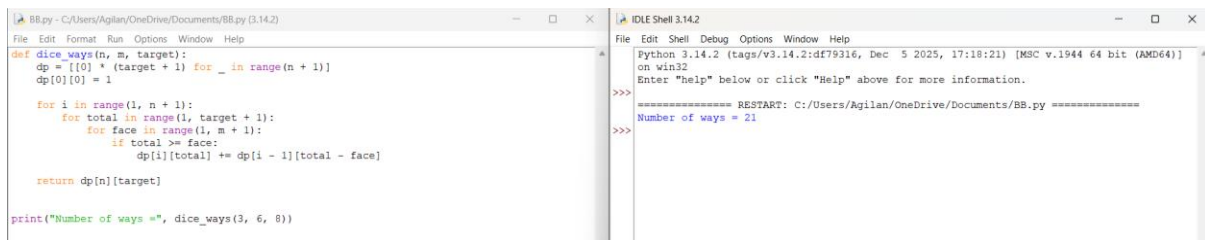
```

```

print("Number of ways =", dice_ways(3, 6, 8))

```

EXECUTION & OUTPUT:



Q10. 4 DICE, 6 FACES, TARGET=10:

```

def dice_ways(n, m, target):

    dp = [[0] * (target + 1) for _ in range(n + 1)]

    dp[0][0] = 1

    for i in range(1, n + 1):

        for total in range(1, target + 1):

            for face in range(1, m + 1):

                if total >= face:

                    dp[i][total] += dp[i - 1][total - face]

    return dp[n][target]

```



```
print("Number of ways =", dice_ways(4, 6, 10))
```

EXECUTION & OUTPUT:



```
BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)
File Edit Format Run Options Window Help
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for total in range(1, target + 1):
            for face in range(1, m + 1):
                if total >= face:
                    dp[i][total] += dp[i - 1][total - face]

    return dp[n][target]

print("Number of ways =", dice_ways(4, 6, 10))

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
***** RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py *****
>>> |
Number of ways = 80
```

Q11. 2 DICE, 4 FACES, TARGET=5:

```
def dice_ways(n, m, target):
```

```
dp = [[0] * (target + 1) for _ in range(n + 1)]
```

$$\text{dp}[0][0] = 1$$

```
for i in range(1, n + 1):
```

```
for total in range(1, target + 1):
```

```
for face in range(1, m + 1):
```

```

if total >= face:

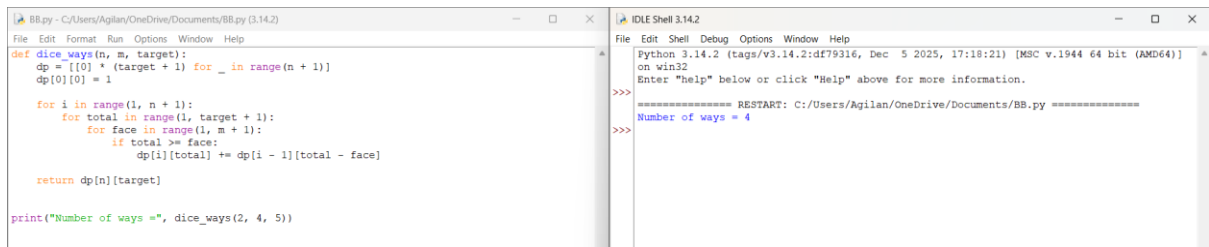
```

```
dp[i][total] += dp[i - 1][total - face]
```

```
return dp[n][target]
```

```
print("Number of ways =", dice_ways(2, 4, 5))
```

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following code:

```
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for total in range(1, target + 1):
            for face in range(1, m + 1):
                if total >= face:
                    dp[i][total] += dp[i - 1][total - face]

    return dp[n][target]

print("Number of ways =", dice_ways(2, 4, 5))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 4
>>>
```

BINOMIAL COEFFICIENT

DICE THROWS-3 DICE, 4 FACES, TARGET=7:

def dice_ways(n, m, target):

 dp = [[0] * (target + 1) for _ in range(n + 1)]

 dp[0][0] = 1

 for i in range(1, n + 1):

 for j in range(1, target + 1):

 for face in range(1, m + 1):

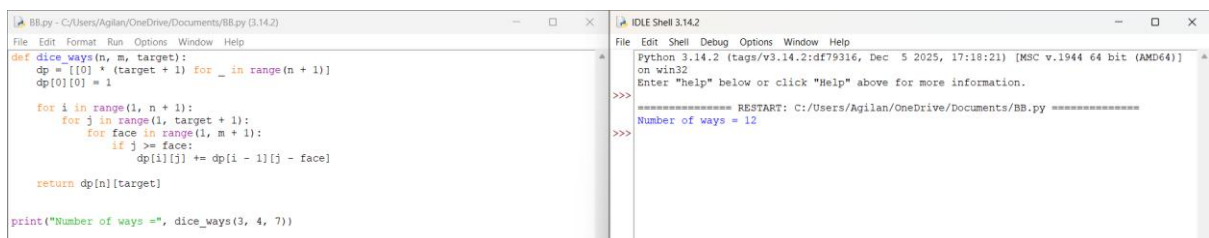
 if j >= face:

 dp[i][j] += dp[i - 1][j - face]

 return dp[n][target]

print("Number of ways =", dice_ways(3, 4, 7))

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'BB.py - C:/Users/Agilan/OneDrive/Documents/BB.py (3.14.2)', contains the following code:

```
def dice_ways(n, m, target):
    dp = [[0] * (target + 1) for _ in range(n + 1)]
    dp[0][0] = 1

    for i in range(1, n + 1):
        for j in range(1, target + 1):
            for face in range(1, m + 1):
                if j >= face:
                    dp[i][j] += dp[i - 1][j - face]

    return dp[n][target]

print("Number of ways =", dice_ways(3, 4, 7))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/BB.py =====
Number of ways = 12
>>>
```

Q2. $C(5,2)$:

```
def binomial(n, r):
```

```
    dp = [[0] * (r + 1) for _ in range(n + 1)]
```

```
    for i in range(n + 1):
```

```
        dp[i][0] = 1
```

```
    for i in range(1, n + 1):
```

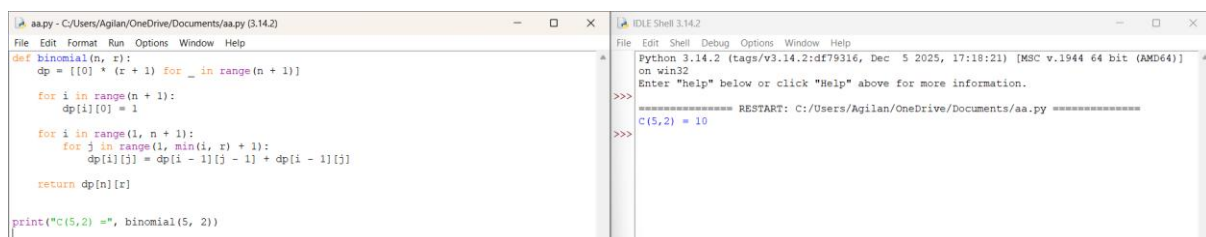
```
        for j in range(1, min(i, r) + 1):
```

```
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
```

```
    return dp[n][r]
```

```
print("C(5,2) =", binomial(5, 2))
```

EXECUTION & OUTPUT:



```
aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)
File Edit Format Run Options Window Help
def binomial(n, r):
    dp = [[0] * (r + 1) for _ in range(n + 1)]
    for i in range(n + 1):
        dp[i][0] = 1
    for i in range(1, n + 1):
        for j in range(1, min(i, r) + 1):
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
    return dp[n][r]
print("C(5,2) =", binomial(5, 2))

IDLE Shell 3.14.2
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
>>>
>>> C(5,2) = 10
>>>
```

$C(8,3)$:

```
def binomial(n, r):
```

```
    dp = [[0] * (r + 1) for _ in range(n + 1)]
```

```
    for i in range(n + 1):
```

```
        dp[i][0] = 1
```

```

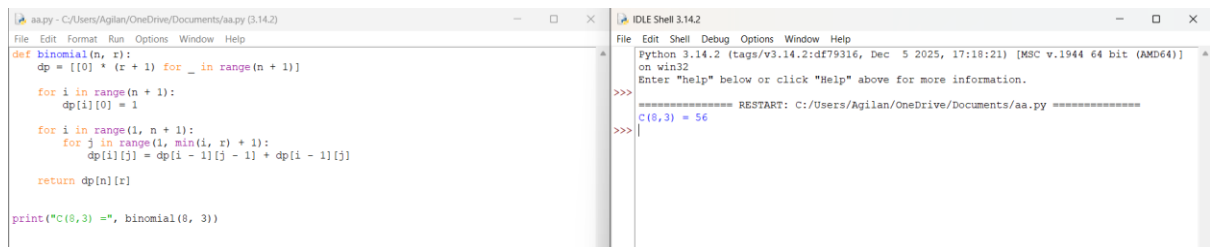
for i in range(1, n + 1):
    for j in range(1, min(i, r) + 1):
        dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

return dp[n][r]

```

```
print("C(8,3) =", binomial(8, 3))
```

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)', contains the following code:

```

def binomial(n, r):
    dp = [[0] * (r + 1) for _ in range(n + 1)]
    for i in range(n + 1):
        dp[i][0] = 1
    for i in range(1, n + 1):
        for j in range(1, min(i, r) + 1):
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
    return dp[n][r]

print("C(8,3) =", binomial(8, 3))

```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====
>>> C(8,3) = 56
>>>

```

C(10,4):

```

def binomial(n, r):
    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):
        dp[i][0] = 1

    for i in range(1, n + 1):
        for j in range(1, min(i, r) + 1):
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

    return dp[n][r]

```

```
print("C(10,4) =", binomial(10, 4))
```

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)', contains the following code:

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
    for i in range(n + 1):  
        dp[i][0] = 1  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
    return dp[n][r]  
  
print("C(10,4) =", binomial(10, 4))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
>>>  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
>>>  
C(10,4) = 210  
>>>
```

C(12,5):

```
def binomial(n, r):
```

```
    dp = [[0] * (r + 1) for _ in range(n + 1)]
```

```
    for i in range(n + 1):
```

```
        dp[i][0] = 1
```

```
    for i in range(1, n + 1):
```

```
        for j in range(1, min(i, r) + 1):
```

```
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]
```

```
    return dp[n][r]
```

```
print("Number of Committees =", binomial(12, 5))
```

EXECUTION & OUTPUT:

The screenshot shows a Python IDE with two windows. The left window, titled 'aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)', contains the following code:

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
  
    for i in range(n + 1):  
        dp[i][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
  
    return dp[n][r]  
  
print("Number of Committees =", binomial(12, 5))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
>>>  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
Number of Committees = 792  
>>>
```

$C(15,6)$:

def binomial(n, r):

dp = [[0] * (r + 1) for _ in range(n + 1)]

for i in range(n + 1):

dp[i][0] = 1

for i in range(1, n + 1):

for j in range(1, min(i, r) + 1):

dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

return dp[n][r]

print("C(15,6) =", binomial(15, 6))

EXECUTION & OUTPUT:

The screenshot shows the same Python IDE as before, but with the code modified to calculate $C(15,6)$. The left window now contains:

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
  
    for i in range(n + 1):  
        dp[i][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
  
    return dp[n][r]  
  
print("C(15,6) =", binomial(15, 6))
```

The right window shows the output:

```
>>>  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
C(15,6) = 5005  
>>>
```

$C(10,3)$:

def binomial(n, r):

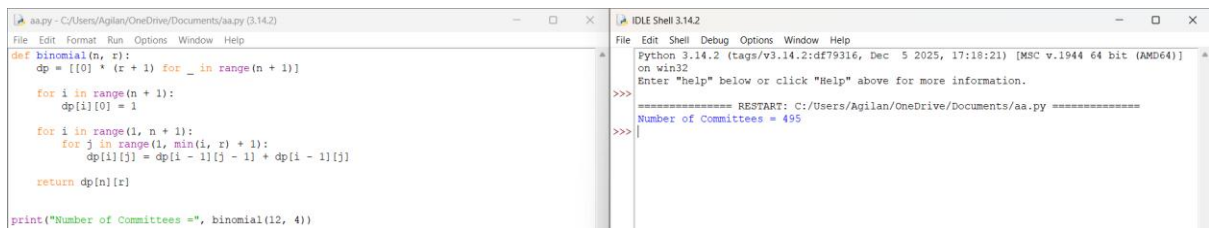
```
for j in range(1, min(i, r) + 1):
```

$$dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]$$

return dp[n][r]

print("Number of Committees =", binomial(12, 4))

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'aa.py - C:/Users/Agilan/OneDrive/Documents/aa.py (3.14.2)', contains the following code:

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
  
    for i in range(n + 1):  
        dp[i][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
  
    return dp[n][r]  
  
print("Number of Committees =", binomial(12, 4))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
Number of Committees = 495  
>>>
```

C(15,5):

def binomial(n, r):

dp = [[0] * (r + 1) for _ in range(n + 1)]

for i in range(n + 1):

dp[i][0] = 1

for i in range(1, n + 1):

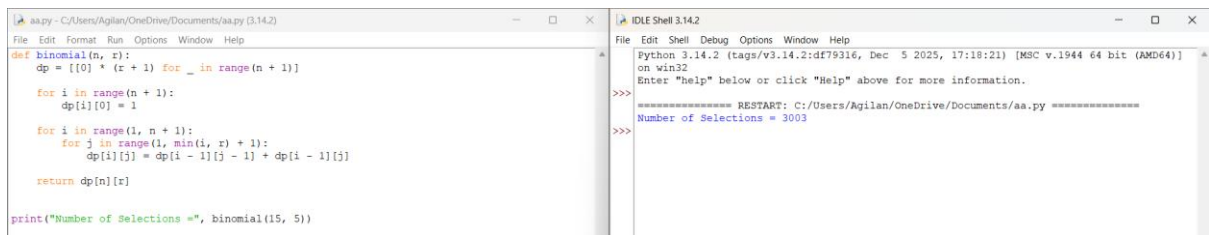
for j in range(1, min(i, r) + 1):

$$dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]$$

return dp[n][r]

print("Number of Selections =", binomial(15, 5))

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'aa.py', contains the following code:

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
  
    for i in range(n + 1):  
        dp[i][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
  
    return dp[n][r]  
  
print("Number of Selections =", binomial(15, 5))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
>>>  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
Number of Selections = 3003  
>>>
```

$C(20,6)$:

def binomial(n, r):

dp = [[0] * (r + 1) for _ in range(n + 1)]

for i in range(n + 1):

dp[i][0] = 1

for i in range(1, n + 1):

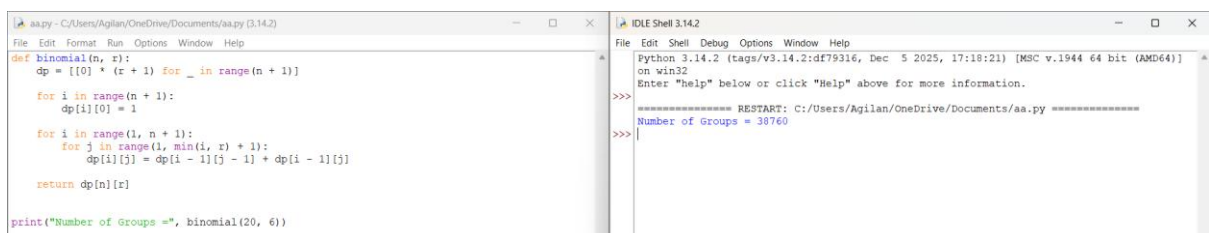
for j in range(1, min(i, r) + 1):

dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

return dp[n][r]

print("Number of Groups =", binomial(20, 6))

EXECUTION & OUTPUT:



The screenshot shows a Python IDE with two windows. The left window, titled 'aa.py', contains the following code:

```
def binomial(n, r):  
    dp = [[0] * (r + 1) for _ in range(n + 1)]  
  
    for i in range(n + 1):  
        dp[i][0] = 1  
  
    for i in range(1, n + 1):  
        for j in range(1, min(i, r) + 1):  
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]  
  
    return dp[n][r]  
  
print("Number of Groups =", binomial(20, 6))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
>>>  
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====  
Number of Groups = 38760  
>>>
```

$C(8,3)$:

```
def binomial(n, r):

    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):

        dp[i][0] = 1

    for i in range(1, n + 1):

        for j in range(1, min(i, r) + 1):

            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

    return dp[n][r]

print("Number of Combinations =", binomial(8, 3))
```

EXECUTION & OUTPUT:

```
def binomial(n, r):
    dp = [[0] * (r + 1) for _ in range(n + 1)]

    for i in range(n + 1):
        dp[i][0] = 1

    for i in range(1, n + 1):
        for j in range(1, min(i, r) + 1):
            dp[i][j] = dp[i - 1][j - 1] + dp[i - 1][j]

    return dp[n][r]

print("Number of Combinations =", binomial(8, 3))
```

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Agilan/OneDrive/Documents/aa.py =====
Number of Combinations = 56
>>>
```

ASSEMBLY LINE SCHEDULING

```
entry = [10, 12]

exit = [18, 7]

line1 = [4, 5, 3, 2]

line2 = [2, 10, 1, 4]
```

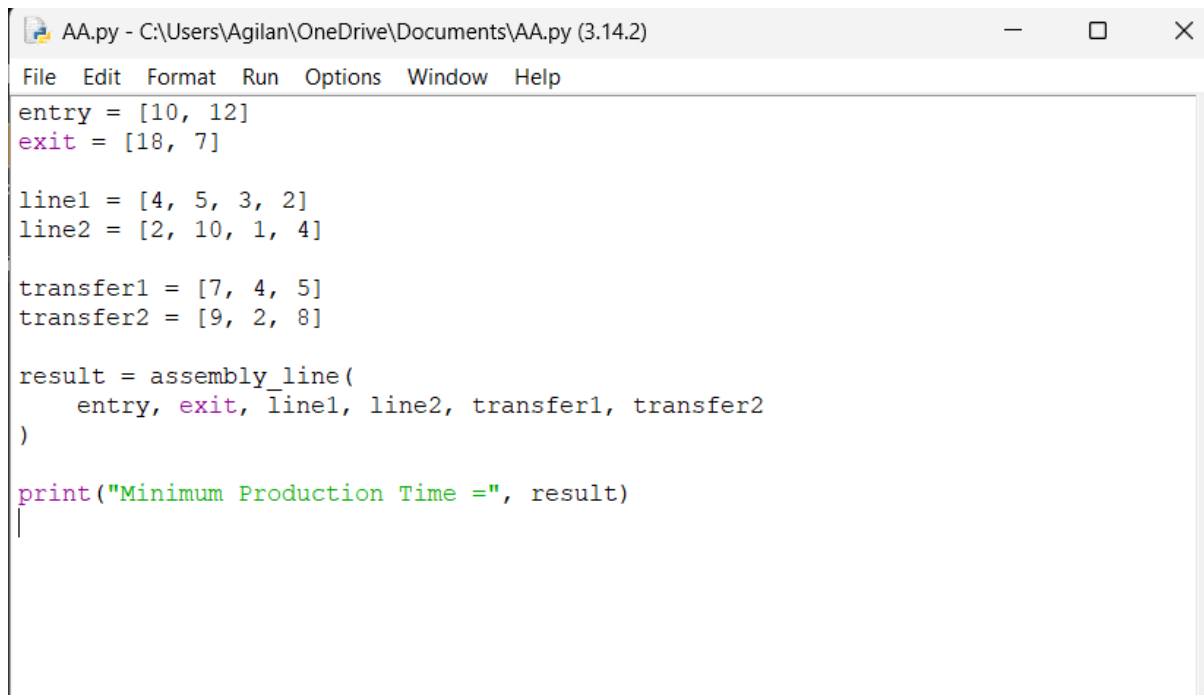
```
transfer1 = [7, 4, 5]
```

```
transfer2 = [9, 2, 8]
```

```
result = assembly_line(  
    entry, exit, line1, line2, transfer1, transfer2  
)
```

```
print("Minimum Production Time =", result)
```

EXECUTION & OUTPUT:

A screenshot of a Python IDE window titled "AA.py - C:\Users\Agilan\OneDrive\Documents\AA.py (3.14.2)". The window has a menu bar with "File", "Edit", "Format", "Run", "Options", "Window", and "Help". The code editor displays the following Python code:

```
entry = [10, 12]  
exit = [18, 7]  
  
line1 = [4, 5, 3, 2]  
line2 = [2, 10, 1, 4]  
  
transfer1 = [7, 4, 5]  
transfer2 = [9, 2, 8]  
  
result = assembly_line(  
    entry, exit, line1, line2, transfer1, transfer2  
)  
  
print("Minimum Production Time =", result)
```